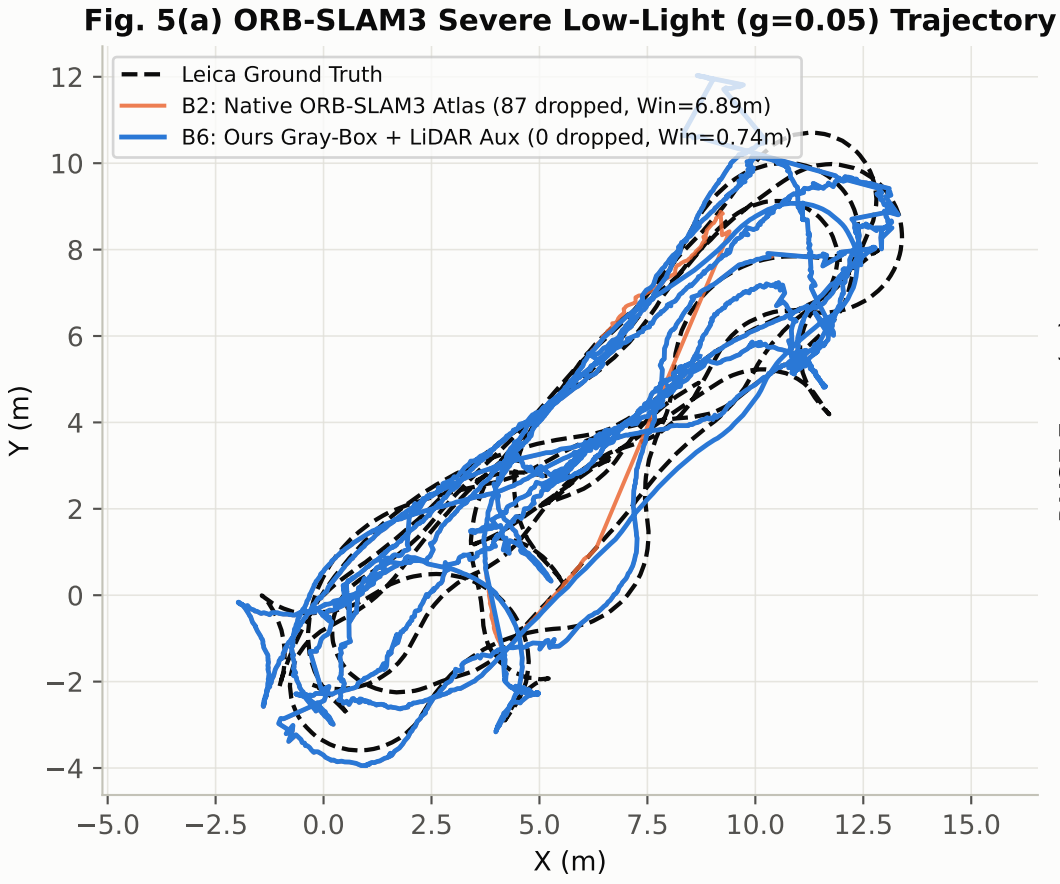
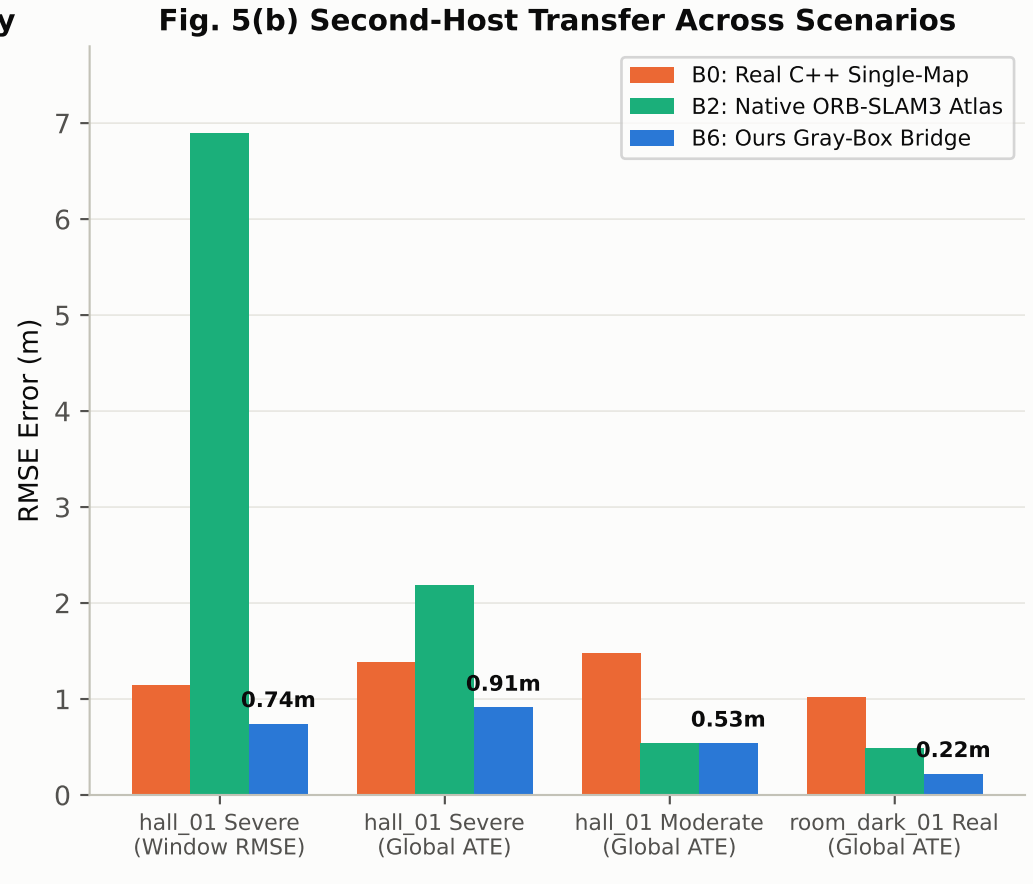

Fig. 5(a) ORB-SLAM3 Severe Low-Light (g=0.05) Trajectory
Leica Ground Truth
B2: Native ORB-SLAM3 Atlas (87 dropped, Win=6.89m)
B6: Ours Gray-Box + LiDAR Aux (0 dropped, Win=0.74m)
Y (m)
X (m)
12
10
8
6
4
2
0
−2
−4
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
12.5
15.0


Fig. 5(b) Second-Host Transfer Across Scenarios
B0: Real C++ Single-Map
B2: Native ORB-SLAM3 Atlas
B6: Ours Gray-Box Bridge
RMSE Error (m)
7
6
5
4
3
2
1
0
0.74m
0.91m
0.53m
0.22m
hall_01 Severe
(Window RMSE)
hall_01 Severe
(Global ATE)
hall_01 Moderate
(Global ATE)
room_dark_01 Real
(Global ATE)